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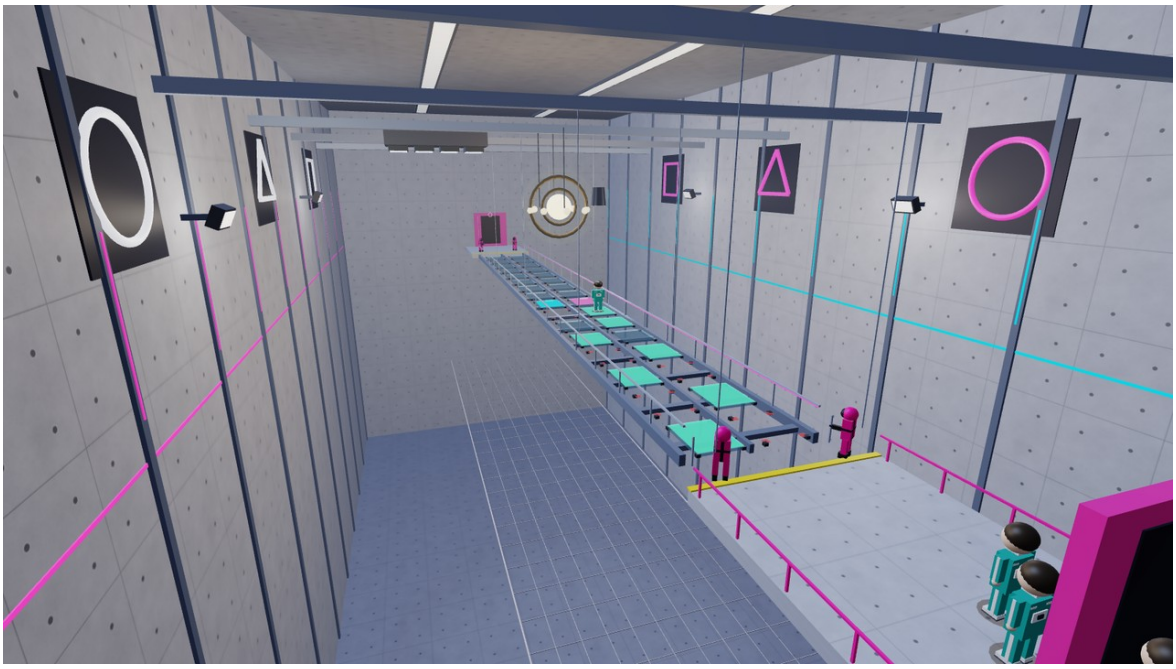
Department of Computer Science and Engineering

CSE 4102: Computer Graphics Laboratory

Final Project Report

Squid Game 3D

Red Light, Green Light & The Glass Bridge



A two-round 3D game made with C++ and modern OpenGL. Every object is built from basic shapes, lit by three kinds of light, Phong shaded and partly texture mapped.

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1 Introduction

This project is a small 3D game based on the TV series *Squid Game*. It has two rounds:

- **Round 1 – Red Light, Green Light.** Walk to the pink finish line. Move only while the giant doll sings. If you move when she turns around, you are out.
- **Round 2 – The Glass Bridge.** Cross 16 pairs of glass panes. In each pair one pane is strong and one breaks. Pick left or right and hope.

What this report shows:

- how every object is built from 7 basic shapes, with real pictures and transform tables (Section 8);
- the three light types – directional, point and spot (Section 9);
- Phong shading, done for every pixel (Section 10);
- texture mapping with three image textures (Section 11);
- animation, game logic and cameras (Sections 12 and 13).

All pictures in this report are real screenshots from the program. The game has a special *capture mode* that sets up each scene and saves the picture automatically.

2 The Game in Pictures

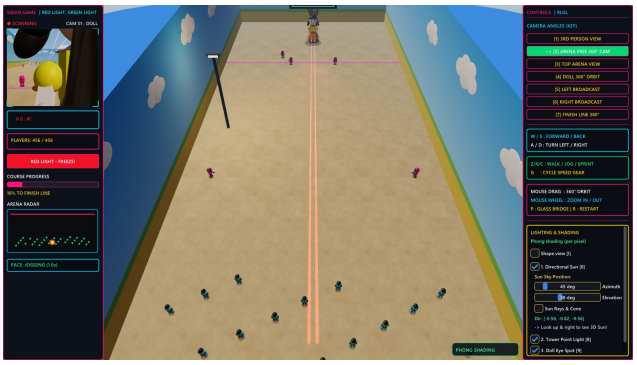
2.1 Round 1: Red Light, Green Light



Figure 1: Start screen. Left panel: live doll camera, timer, player count, progress bar and a mini radar. Right panel: camera buttons, controls and the lighting panel.



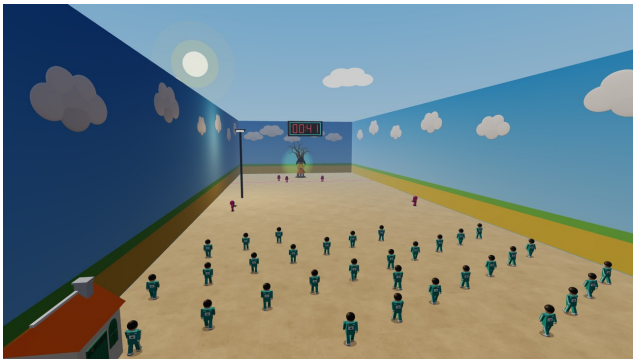
(a) Green light – everyone walks.



(b) Red light – the doll turns and scans with red lasers.



(c) Player 456 crossed the finish line.



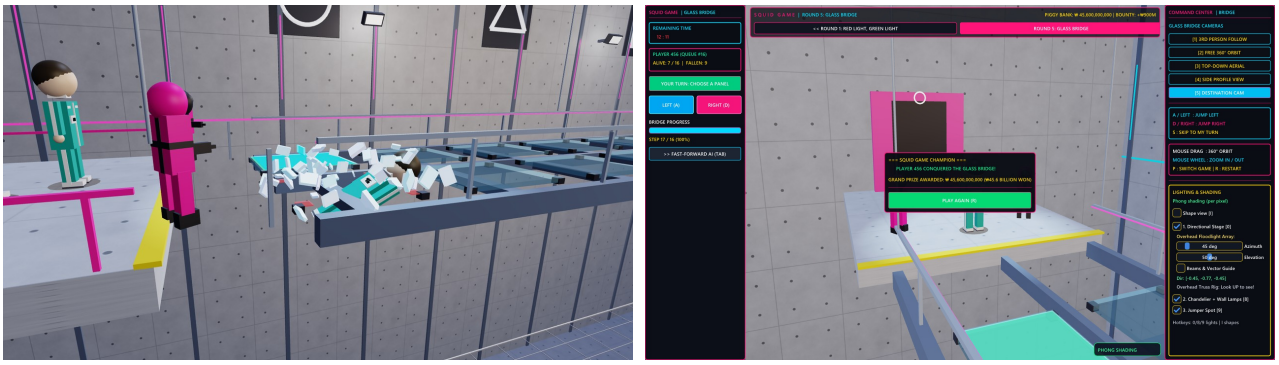
(d) The whole arena, seen from behind the start.

Figure 2: Round 1 while playing.

2.2 Round 2: The Glass Bridge



Figure 3: Player 456's turn. Green panes are proven safe, the red marks show broken panes. The cyan and pink panes in front are the two choices: press `A` for left or `D` for right.



(a) Wrong choice: the pane breaks into 55 pieces. **(b)** Player 456 reached the finish platform.

Figure 4: Round 2 while playing.

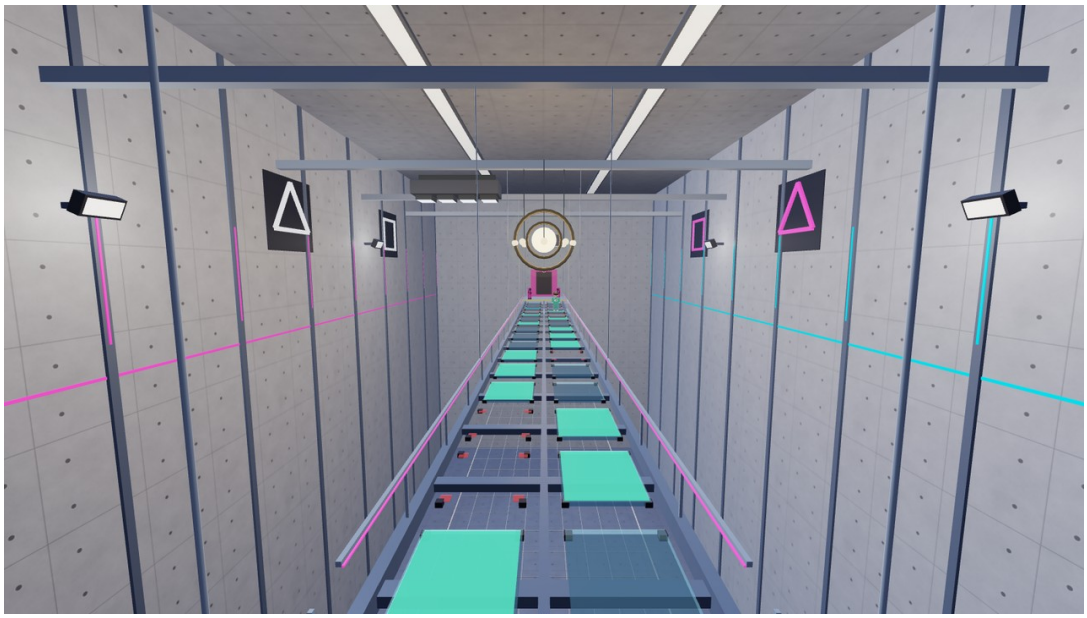


Figure 5: The glass bridge hall, made brighter in this version: light concrete walls on both sides, four wall lamps (extra point lights), two ceiling light strips, neon symbols and LED bars.

3 Tools Used

Item	What we used
Language	C++17, compiled with MinGW-w64 g++
Graphics	OpenGL 3.3 Core Profile with GLSL 330 shaders
Window and input	GLFW 3
OpenGL loader	GLAD
Math	GLM (vectors and matrices)
User interface	Dear ImGui (side panels, buttons, overlays)
Images	stb_image (loads the PNG textures)
Sound	Windows multimedia API (song + generated sound effects)
3D models	None imported – every object is built in code from basic shapes

4 How to Play

Key	Action
Both rounds	
`P`	Switch between Round 1 and Round 2
`R`	Restart the round
`0 / 8 / 9`	Turn the directional / point / spot light on or off
`I`	Shape view: color every part by its basic shape
`Mouse drag, wheel`	Orbit the camera, zoom
Round 1	
`W / S`	Walk forward / backward
`A / D, Q / E`	Turn left / right, step sideways
`Z / X / C (or Shift)`	Walk / jog / sprint
`1 -- 7`	Camera modes
Round 2	
`A / D`	Jump to the left / right pane
`S`	Skip to Player 456's turn
`Tab`	Fast-forward the other players
`1 -- 5`	Camera modes

5 Program Structure

The code is split into small folders, one job each:

Folder	What it does
`main_modern.cpp`	Main loop, keyboard/mouse, lights setup, user interface, capture mode
`engine/`	OpenGL building blocks: Shader, VAO, VBO, EBO, Mesh, Texture, basic shapes, matrix stack
`shaders/`	`default.vert` and `default.frag` (Phong lighting)
`render/`	Light fixture models, the three textures
`contestants/, guards/, doll/`	The characters
`environment/`	Round 1 arena: sky, sand, walls, houses, clock
`glassbridge/`	Round 2: bridge, hall, platforms, glass shards, cameras, game rules
`camera/, audio/`	Round 1 cameras, sound
`core/Config.h`	All colors, sizes, speeds and timers in one place
`tools/`	Shape showcase (report figures), texture generator, figure script

Every frame the program does the same five steps:

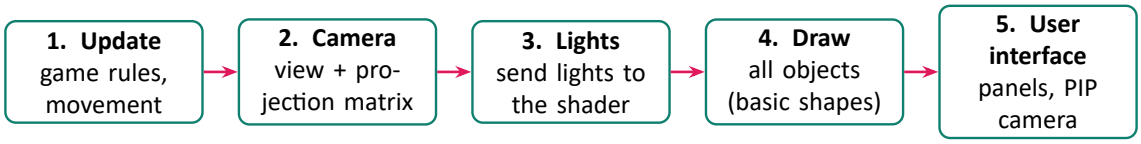


Figure 6: What happens in one frame (about 60 times per second).

World size. All sizes are in game units. Round 1 arena: x from -24 to 24, z from -75 to 75, walls 22 high. Start line at $z = -52$, finish line at $z = 45$, doll at $(0, 0, 58)$, 2.6 times bigger than normal. Round 2: 16 glass steps, the first at $z = 3$, one step every 3.3 units, at height $y = 16$.

6 Basic Shapes

The whole game is built from only seven basic shapes. Figure 7 shows each one, rendered by the program itself. The thin dark lines show the triangles the GPU actually draws.

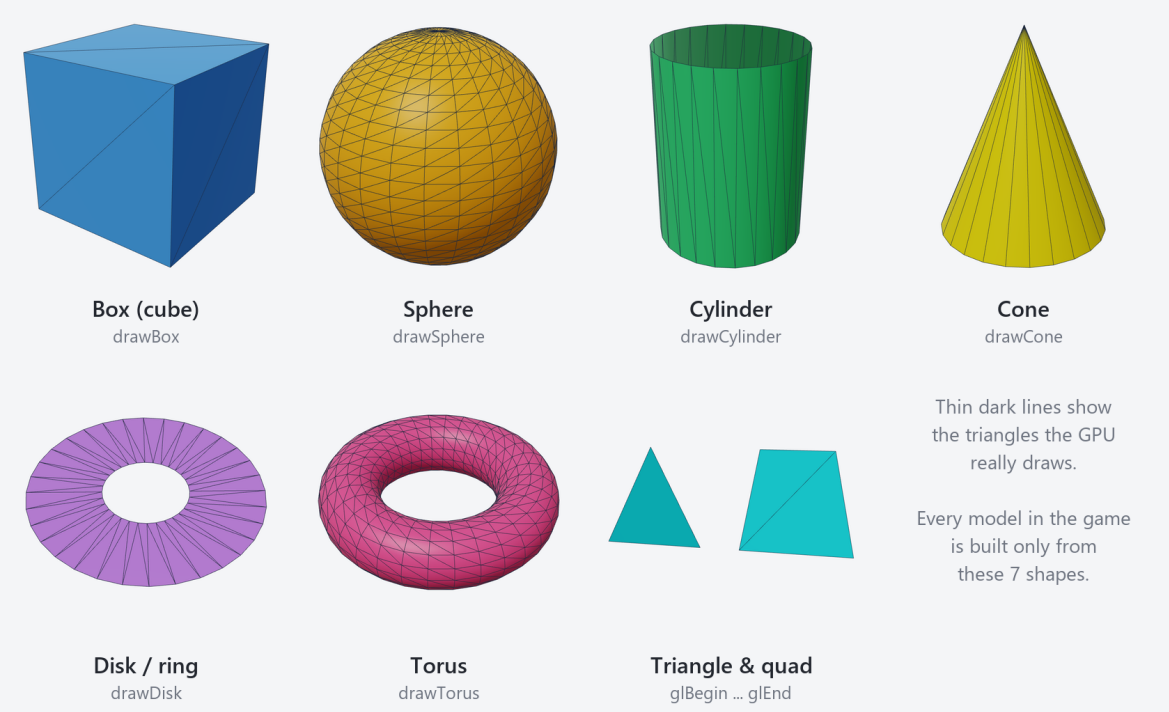


Figure 7: The seven basic shapes and the function that draws each one.

Shape	Settings	How it is made	Used for (examples)
Box	width, height, depth	6 faces × 4 corners = 24 vertices, 12 triangles	bodies, legs, walls, glass panes, rails
Sphere	radius	24 rings × 24 slices = 1152 triangles	heads, eyes, hands, clouds, bulbs
Cylinder	bottom radius, top radius, height	a ring of side rectangles (no lids)	tree branches, neck, cables
Cone	radius, height	a cylinder whose top radius is 0	(cone beams)
Disk	inner radius, outer radius	flat ring of triangles	doll's collar, lens, sleeve cuffs
Torus	tube radius, ring radius	a tube bent into a ring	chandelier rings, circle symbols
Polygon	any corner points	triangles / quads typed in by hand	roofs, sand floor, skirt, sky

Table 1: The basic shapes. A box is always made as a $1 \times 1 \times 1$ cube and stretched with *scale*; a sphere is always radius 1 and scaled to the wanted size.

7 Transformations

Each shape is moved into place with three transformations. In 3D we use 4×4 matrices:

$$T = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad S = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Translate** (T) moves a part. **Scale** (S) stretches it. **Rotate** (R_x, R_y, R_z) turns it.

- A part's final matrix is **parent** × **T** × **R** × **S**. So the hand follows the arm, and the arm follows the body.
- `pushMatrix()` saves the current matrix and `popMatrix()` brings it back. This keeps each part's changes from leaking into the next part.

Example – one leg of Player 456 (from `Contestants.cpp`):

```
glPushMatrix();
glTranslatef(side * 0.15f, 0.62f, 0.0f); // move to the hip joint
glRotatef(legAngle, 1.0f, 0.0f, 0.0f); // swing the leg forward/back (walk)
glTranslatef(0.0f, -0.279f, 0.0f); // the leg hangs below the hip
glScalef(0.18f, 0.558f, 0.18f); // stretch the unit cube into a leg
glutSolidCube(1.0);
glPopMatrix();
```

Note: these old-style function names are kept on purpose, but they run on modern OpenGL. Our `GLCompat` layer keeps its own matrix stack (GLM) and draws the shape with VAO/VBO buffers and shaders.

8 Objects and the Shapes They Are Made Of

For every object below, the left picture is the finished model. The right picture is the *same code* drawn in shape view: every part is painted with the color of its basic shape and pulled a little away from the center so the parts can be seen. Colors: Box Sphere Cylinder Cone Disk Torus Polygon.

How to read the tables. Positions are in the object's own coordinates, before the whole object is placed in the world. "±" means a left and a right copy. Rotations are written as axis + angle, e.g. $R_x\,25^\circ$ turns 25° around the x axis.

Object	Box	Sphere	Cyl.	Disk	Torus	Poly.	Line	Total
Player 456 / contestant	49	4						53
Pink guard (Round 1)	19	4						23
Bridge guard (Round 2)	9	2			1			12
Young-hee doll (body + head)	50	70	4	5		18		147
of which the head	10	40	1					51
Dead tree		157	156					313
Village house	21	1			3	2	2	29
Countdown clock	55							55
Glass bridge (3 steps shown)	48							48
Start platform	17							17
Light fixtures (5 kinds)	21	9	1	1	2			34

Table 2: Number of parts in each object, counted by the program while drawing it.

8.1 Player 456 and the other contestants

The same model is used for every contestant in both rounds; only the number on the chest and back changes. **Placed in the world with:** $T(x, 0, z) \cdot R_y(\text{facing}) \cdot S(1.15)$ (other contestants $S(1.10)$).

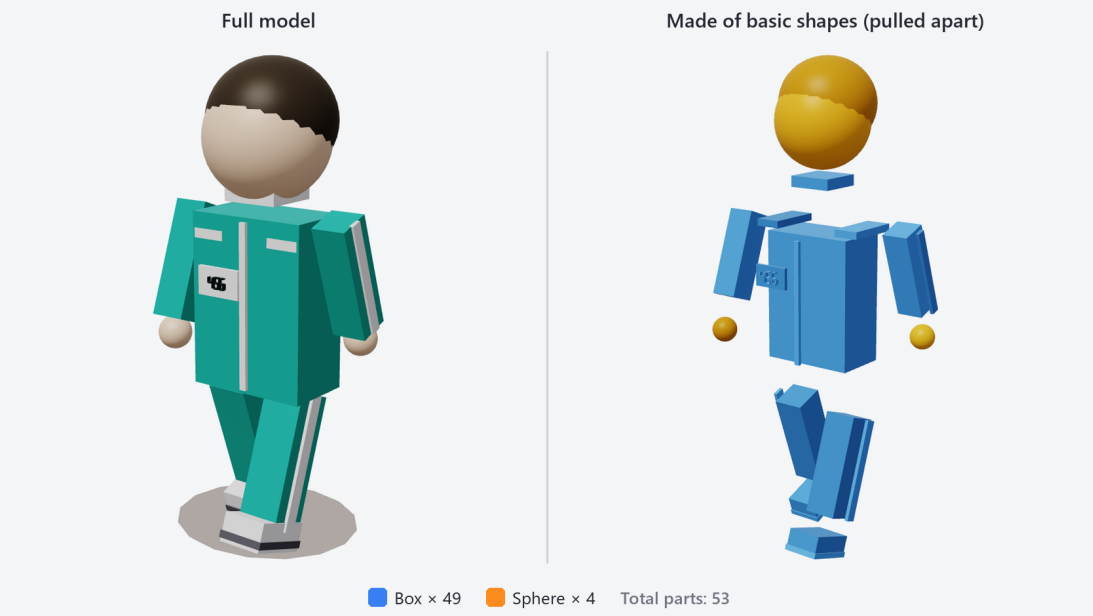


Figure 8: Player 456: 49 boxes and 4 spheres.

Part	Shape	Scale / size	Translate (position)	Rotate
Legs $\times 2$	Box	$0.18 \times 0.56 \times 0.18$	$(\pm 0.15,\ 0.34,\ 0)$	R_x swing at hip (y=0.62)
Leg stripes $\times 2$	Box	$0.02 \times 0.56 \times 0.06$	$(\pm 0.245,\ 0.34,\ 0)$	moves with leg
Shoes $\times 2$	Box	$0.19 \times 0.12 \times 0.28$	$(\pm 0.15,\ 0.04,\ 0.04)$	moves with leg
Shoe soles $\times 2$	Box	$0.20 \times 0.03 \times 0.29$	$(\pm 0.15,\ 0.01,\ 0.04)$	moves with leg
Jacket (body)	Box	$0.52 \times 0.72 \times 0.30$	$(0,\ 0.98,\ 0)$	–
Zipper	Box	$0.03 \times 0.68 \times 0.02$	$(0,\ 0.98,\ 0.16)$	–
Shoulder stripes $\times 2$	Box	$0.14 \times 0.04 \times 0.31$	$(\pm 0.18,\ 1.25,\ 0)$	–
Chest number patch	Box	$0.20 \times 0.12 \times 0.02$	$(-0.12,\ 1.07,\ 0.16)$	–
Back number patch	Box	$0.32 \times 0.18 \times 0.02$	$(0,\ 1.05,\ -0.16)$	$R_y\ 180°$
Number “456” (front + back)	Box $\times 30$	7-segment bars, 0.021×0.005 (front)	on the patches	–
Arms $\times 2$	Box	$0.15 \times 0.49 \times 0.15$	$(\pm 0.335,\ 1.10,\ 0)$	R_x swing at shoulder (y=1.34)
Arm stripes $\times 2$	Box	$0.02 \times 0.49 \times 0.05$	$(\pm 0.41,\ 1.10,\ 0)$	moves with arm
Hands $\times 2$	Sphere	radius 0.07	$(\pm 0.335,\ 0.80,\ 0)$	moves with arm
Collar	Box	$0.24 \times 0.06 \times 0.24$	$(0,\ 1.42,\ 0)$	–
Face	Sphere	radius 0.26	$(0,\ 1.64,\ 0)$	–
Hair	Sphere	radius 0.26, $S(1.02, 0.95, 1.02)$	$(0,\ 1.70,\ -0.03)$	–

8.2 Pink guard (Round 1)

Five guards stand around the arena. The mask symbol shows the rank: circle, triangle or square. **Placed with:** $T(x, 0, z) \cdot R_y(\text{facing}) \cdot S(1.2)$. When a guard shoots, it turns to the target: facing $=$ atan2$(x_t - x_g,\ z_t - z_g)$.

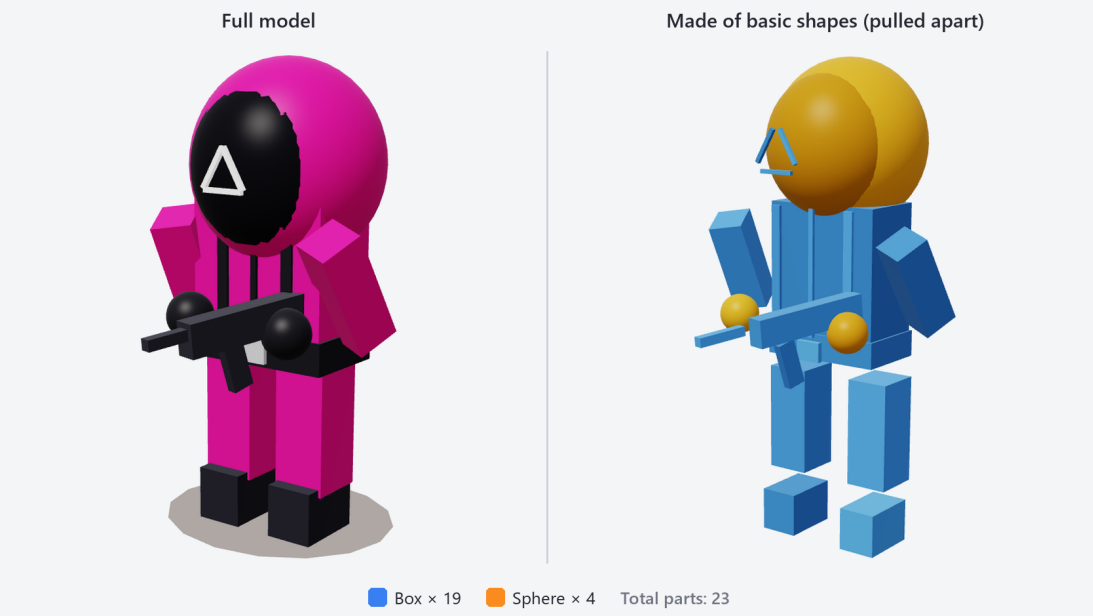


Figure 9: Triangle guard: 19 boxes and 4 spheres.

Part	Shape	Scale / size	Translate (position)	Rotate
Boots $\times 2$	Box	$0.18 \times 0.20 \times 0.28$	$(\pm 0.16,\ 0.10,\ 0.03)$	–
Legs $\times 2$	Box	$0.20 \times 0.50 \times 0.22$	$(\pm 0.16,\ 0.42,\ 0)$	–
Body	Box	$0.55 \times 0.60 \times 0.32$	$(0,\ 0.95,\ 0)$	–
Zipper	Box	$0.024 \times 0.58 \times 0.015$	$(0,\ 0.95,\ 0.165)$	–
Belt + buckle	Box $\times 2$	$0.57 \times 0.12 \times 0.34$; $0.12 \times 0.09 \times 0.025$	$(0,\ 0.70,\ 0)$; $(0,\ 0.70,\ 0.175)$	–
Shoulder straps $\times 2$	Box	$0.045 \times 0.50 \times 0.015$	$(\pm 0.14,\ 0.98,\ 0.163)$	–
Neck collar	Box	$0.34 \times 0.09 \times 0.32$	$(0,\ 1.22,\ 0)$	–
Left / right arm	Box	$0.16 \times 0.45 \times 0.16$	$(-0.35,\ 0.95,\ 0.08)$ / $(0.35,\ 0.95,\ 0.10)$	$R_x 25^\circ R_y 20^\circ$ / $R_x 35^\circ R_y -25^\circ$
Gloves $\times 2$	Sphere	radius 0.09	$(-0.25,\ 0.80,\ 0.22)$, $(0.20,\ 0.80,\ 0.25)$	–
Hood	Sphere	$S(0.36, 0.38, 0.36)$	$(0,\ 1.35,\ -0.04)$	–
Black mask	Sphere	$S(0.30, 0.32, 0.28)$	$(0,\ 1.35,\ 0.06)$	–
Triangle symbol	Box $\times 3$	base 0.16×0.018, sides 0.018×0.165	mask $+(0, 0, 0.285)$; sides at $x = \pm 0.042$	sides $R_z \mp 30^\circ$
Gun body	Box	$0.08 \times 0.12 \times 0.55$	$(0.12,\ 0.85,\ 0.35)$	$R_y -15^\circ\ R_x 5^\circ$
Barrel, magazine	Box $\times 2$	$0.04 \times 0.04 \times 0.22$; $0.04 \times 0.20 \times 0.08$	gun $+(0, 0.02, 0.32)$; gun $+(0, -0.12, 0.08)$	magazine $R_x 12^\circ$
Muzzle flash (only when shooting)	Sphere	radius 0.14	gun $+(0,\ 0.02,\ 0.48)$	–

Circle guards use one Torus (tube 0.016, ring 0.088) and square guards use four boxes instead of the triangle.

8.3 Young-hee, the giant doll

The doll stands with her arms around the tree. Only her head turns. **Placed with:** $T(0, 0, 58) \cdot S(2.6)$; the head adds $T(0, 2.50, 0) \cdot R_y(\text{headAngle})$.

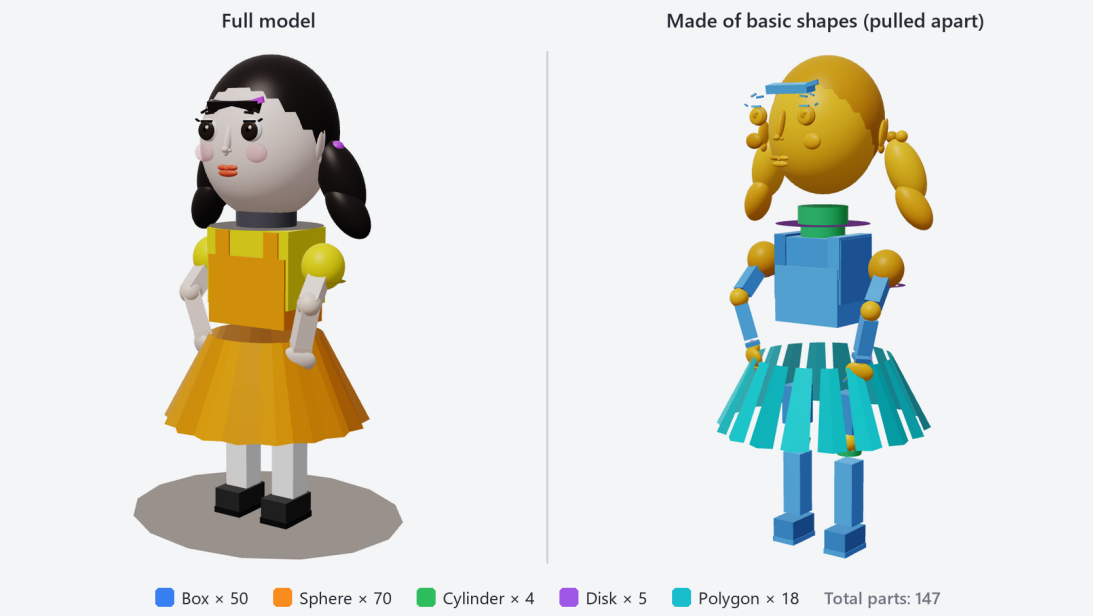


Figure 10: The whole doll: 147 parts. The skirt is 18 hand-made polygon pleats.

Part	Shape	Scale / size	Translate (position)	Rotate
Neck	Cylinder	radius 0.18, height 0.28	$(0,\ 2.05,\ 0)$	$R_x\ 90°$
White collar	Disk	radius $0.18 \rightarrow 0.38$	$(0,\ 1.96,\ 0)$	$R_x\ 90°$
Yellow shirt	Box	$0.64 \times 0.50 \times 0.44$	$(0,\ 1.70,\ 0)$	–
Orange dress top	Box	$0.60 \times 0.45 \times 0.45$	$(0,\ 1.55,\ 0.01)$	–
Dress straps $\times 2$	Box	$0.11 \times 0.32 \times 0.46$	$(\pm 0.19,\ 1.78,\ 0)$	–
Pleated skirt	Polygon $\times 18$	top radius 0.38, bottom 0.65, height 0.58	$(0,\ 1.25,\ 0)$	pleat i at $i \times 20°$
Sleeves $\times 2$	Sphere + Disk	$S(0.14, 0.15, 0.14)$; ring $0.08 \rightarrow 0.135$	shoulder $(\pm 0.44,\ 1.72,\ 0)$	cuff $R_x\ 90°$
Upper arms $\times 2$	Box	$0.12 \times 0.30 \times 0.12$	shoulder $+(0, -0.15, 0)$	$R_x -46° R_y \pm 16° R_z \pm 6°$
Elbows $\times 2$	Sphere + Box	radius 0.078; ring 0.082×0.02	shoulder $+(0, -0.30, 0)$	follows upper arm
Forearms $\times 2$	Box	$0.115 \times 0.29 \times 0.115$	elbow $+(0, -0.15, 0)$	$R_x 58° R_y \pm 34° R_z \mp 8°$
Wrists $\times 2$	Box + Sphere + Disk	cuff 0.12×0.035; ball r 0.052; hole r 0.024	elbow $+(0, -0.29, 0)$	hole $R_y \mp 90°$
Palms $\times 2$	Sphere	$S(0.11, 0.085, 0.045)$	wrist $+(0, -0.08, 0.015)$	$R_y \mp 24° R_x 12°$
Fingers $\times 8$, thumbs $\times 2$	2 Sphere + 2 Box each	joints r 0.010–0.014	finger f at $x = -0.038 + 0.025f$	curl $R_x 16°$
Thighs $\times 2$	Box	$0.18 \times 0.32 \times 0.18$	$(\pm 0.19,\ 0.85,\ 0)$	–
Knees $\times 2$	Sphere	radius 0.09	$(\pm 0.19,\ 0.68,\ 0)$	–
Sock tops $\times 2$	Cylinder	radius $0.11 \rightarrow 0.10$, height 0.05	$(\pm 0.19,\ 0.64,\ 0)$	$R_x\ 90°$
Socks $\times 2$	Box	$0.17 \times 0.52 \times 0.17$	$(\pm 0.19,\ 0.36,\ 0)$	–
Shoes + soles $\times 2$	Box $\times 2$	$0.20 \times 0.16 \times 0.30$; $0.21 \times 0.04 \times 0.31$	$(\pm 0.19,\ 0.09,\ 0.04)$; y=0.02	–

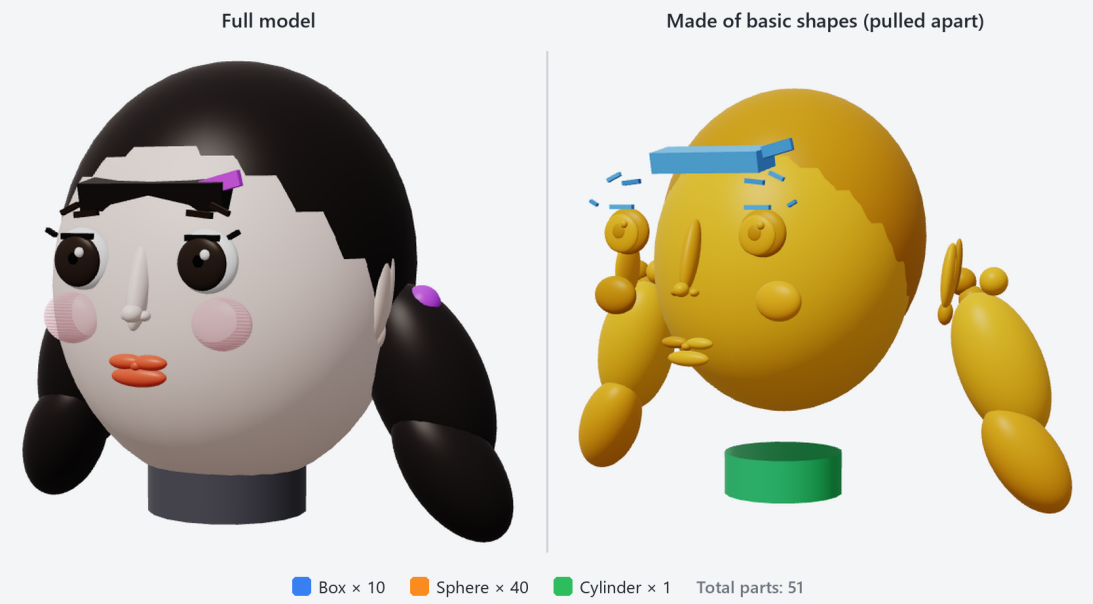


Figure 11: The doll's head on its own: 51 parts, mostly squashed spheres.

Part	Shape	Scale / size	Translate (position)	Rotate
Neck joint ring	Cylinder	radius 0.20, height 0.14	$(0,\ -0.42,\ 0)$	$R_x\ 90°$
Face	Sphere	$S(0.44, 0.50, 0.42)$	$(0,\ 0,\ 0)$	–
Nose bridge, tip	Sphere $\times 2$	$S(0.026, 0.115, 0.034)$; $S(0.024, 0.022, 0.024)$	$(0, 0.03, 0.408)$; $(0, -0.04, 0.444)$	bridge $R_x -18°$
Nostrils $\times 2$	Sphere	$S(0.014, 0.012, 0.016)$	$(\pm 0.023,\ -0.046,\ 0.430)$	–
Lips (4 parts)	Sphere $\times 4$	upper $S(0.046, 0.018, 0.019)$, lower $S(0.072, 0.024, 0.022)$	$(\pm 0.030, -0.152, 0.422)$, $(0, -0.185, 0.418)$	–
Blush $\times 2$ (32% see-through)	Sphere	$S(0.075, 0.060, 0.034)$	$(\pm 0.20,\ -0.06,\ 0.392)$	–
Eyes $\times 2$	4 Sphere + 2 Box each	white $S(0.078, 0.072, 0.030)$, iris, pupil, shine, lashes	$(\pm 0.165,\ 0.078,\ 0.395)$	lash wing $R_z \pm 28°$
Eyebrows $\times 2$	Box $\times 2$ each	0.062×0.014; 0.050×0.012	$(\pm 0.150, 0.182, 0.402)$; $(\pm 0.198, 0.193, 0.397)$	$R_z \mp 6°$; $R_z \mp 24°$
Hair	Sphere	$S(0.455, 0.495, 0.435)$	$(0,\ 0.06,\ -0.05)$	–
Fringe (bangs)	Box	$0.38 \times 0.065 \times 0.11$	$(0,\ 0.23,\ 0.32)$	$R_x 12°$
Hair clip	Box	$0.10 \times 0.035 \times 0.016$	$(0.18,\ 0.25,\ 0.36)$	$R_z 14°$
Ears $\times 2$	Sphere $\times 4$ each	main $S(0.024, 0.105, 0.055)$	$(\pm 0.42,\ -0.02,\ 0.02)$	$R_y \pm 14° R_z -6°$
Pigtails $\times 2$	Sphere $\times 6$ each	4 beads $r\ 0.046$; hair $S(0.14, 0.26, 0.14)$, $S(0.11, 0.20, 0.11)$	$(\pm 0.44,\ -0.07,\ -0.10)$	$R_z \pm 20°$, $R_z \pm 32°$
Red-light lasers $\times 2$	Cylinder	radius $0.010 \rightarrow 0.07$, length 34	from each eye, along $+z$	follows the head

8.4 The dead tree

The tree is made by a small *recursive* function: a branch draws itself, then two smaller branches at its tip, and so on. Every branch is a tapered cylinder with a sphere "knuckle" at the end, so joints have no gaps. The bark is a texture (Section 11). **Placed with:** $T(0, 0, 59.85)$, just behind the doll.

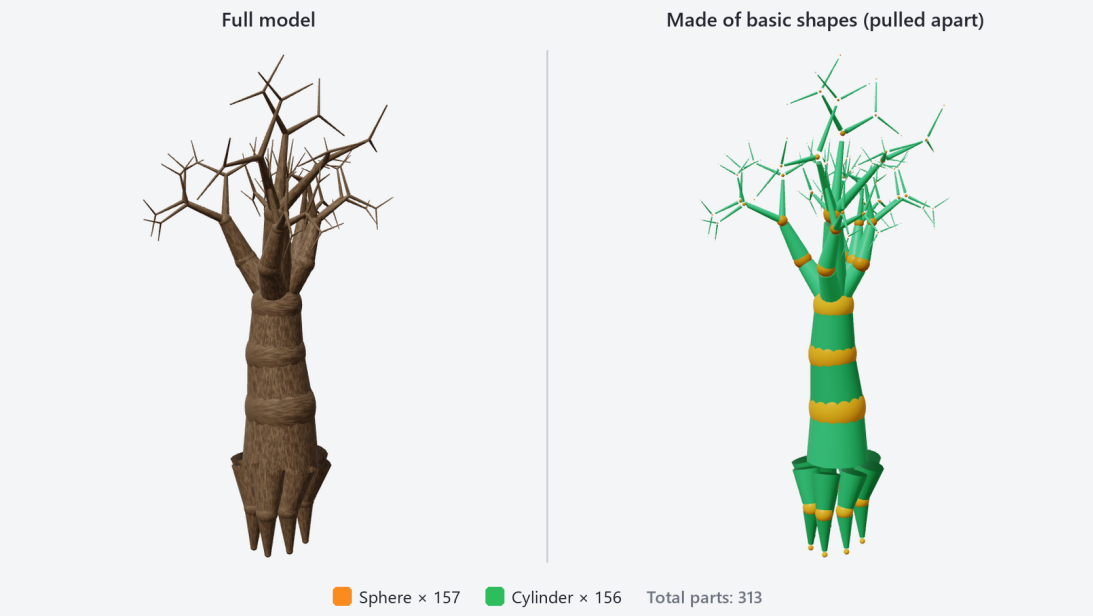


Figure 12: The tree: 156 cylinders and 157 spheres.

Part	Shape	Scale / size	Translate (position)	Rotate
Roots $\times 8$	2 Cylinder + 2 Sphere each	radius $0.65 \to 0.38 \to 0.12$, length $1.6 + 1.6$	$(1.3\cos\theta,\ 0.25,\ 1.3\sin\theta)$, $\theta = i \cdot 45°$	$R_y\theta\ R_x 75°$, then $R_x 18°$
Trunk part 1	Cylinder	radius $1.45 \to 1.25$, height 2.2	$(0,\ 0,\ 0)$	$R_x -90°$ (stand up)
Trunk part 2	Cylinder	radius $1.25 \to 1.05$, height 2.0	on top of part 1	$R_x 3.5° R_y -4°$
Trunk part 3	Cylinder	radius $1.05 \to 0.88$, height 1.8	on top of part 2	$R_x -4.5° R_y 5°$
Fork knuckle	Sphere	radius 0.92	top of trunk (≈ 6 high)	–
Big branches $\times 5$	2 Cylinder + 3 twigs each	base radius 0.50–0.56, length 3.2–3.6	at the fork	R_z: $-48°, 48°, -135°, 135°, 180°$; R_x: $28°$–$38°$
Top spire	Cylinder + 3 twigs	radius $0.65 \to 0.42$, length 3.5	at the fork	$R_x 8° R_y 12°$
Twigs (recursive)	Cylinder + Sphere	each level: length $\times 0.70$, radius $\times 0.45$	at the parent's tip	$R_y \pm (26°$–$40°)$, $R_x \pm (35°$–$70°)$

8.5 Village house

One house model is drawn 9 times: five in a row behind the start ($z = -67$) and two on each side wall ($x = \pm 22$, turned $\pm 90°$). **Placed with:** $T(x, 0, z) \cdot R_y(\text{angle})$.

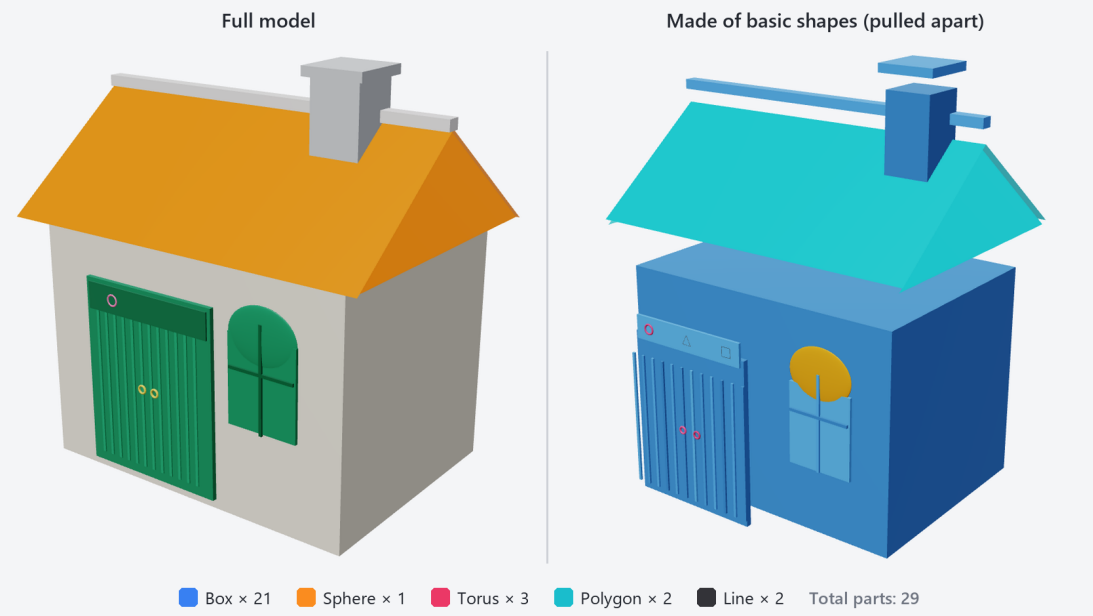


Figure 13: Village house: 29 parts. The roof is made of hand-made triangles and quads.

Part	Shape	Scale / size	Translate (position)	Rotate
Walls (body)	Box	$4.4 \times 3.2 \times 3.2$	$(0,\ 1.6,\ 0)$	–
Roof ends $\times 2$	Polygon (triangle)	base 3.7, height 1.4	$x = \pm 2.45$, from $y = 3.2$	normal $(\pm 0.71, 0.71, 0)$
Roof sides $\times 2$	Polygon (quad)	5.0 long, 1.85 deep, 1.4 rise	front / back, from $y = 3.2$	normal $(0, 0.71, \pm 0.71)$
Roof ridge	Box	$5.0 \times 0.14 \times 0.18$	$(0,\ 4.65,\ 0)$	–
Chimney + cap	Box $\times 2$	$0.65 \times 1.4 \times 0.65$; $0.80 \times 0.12 \times 0.80$	$(1.3,\ 4.4,\ 0.2)$; $(1.3,\ 5.15,\ 0.2)$	–
Door frame	Box	$1.95 \times 2.45 \times 0.08$	$(-0.55,\ 1.35,\ 1.62)$	–
Door bars $\times 12$	Box	$0.04 \times 2.0 \times 0.03$	$x = -1.35$ to 0.19, every 0.14	–
Sign above door	Box	$1.85 \times 0.35 \times 0.04$	$(-0.55,\ 2.35,\ 1.66)$	–
Symbols ◯ △ □	Torus + 2 line loops	ring 0.07; triangle/square 0.14 wide	on the sign	–
Door rings $\times 2$	Torus	tube 0.015, ring 0.05	$(-0.65,\ 1.30,\ 1.68)$, $(-0.45,\ 1.30,\ 1.68)$	–
Window + round top	Box + Sphere	$0.95 \times 1.15 \times 0.04$; $S(0.48, 0.38, 0.04)$	$(1.15,\ 1.8,\ 1.64)$; $+(0, 0.58, 0)$	–
Window bars $\times 2$	Box	0.04×1.35; 0.90×0.04	window $+(0, 0.08, 0.03)$, $+(0, 0.15, 0.03)$	–

8.6 Countdown clock

A digital clock hangs above the finish line and shows the time on both sides. Each digit is drawn with up to 7 thin boxes ("7-segment" style); a small table says which bars are on for each digit 0–9. **Placed with:** $T(0, 17.5, 42) \cdot S(2)$.

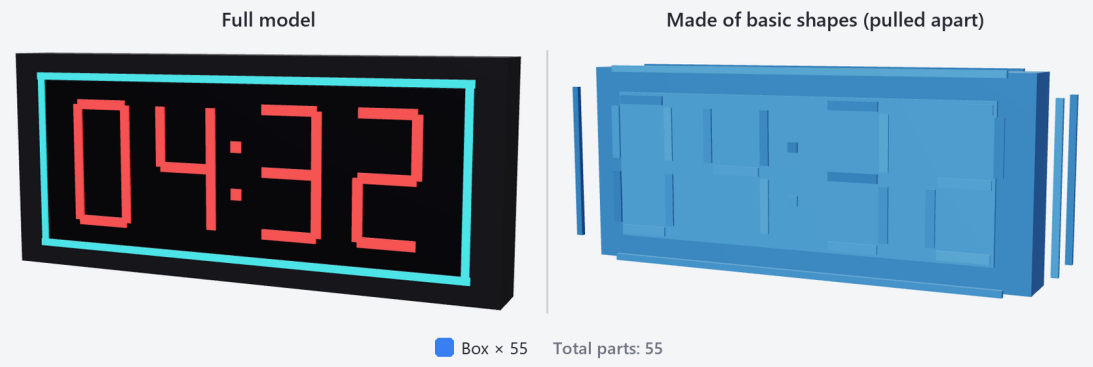


Figure 14: Clock showing 04:32: 55 boxes.

Part	Shape	Scale / size	Translate (position)	Rotate
Outer frame	Box	$5.6 \times 2.4 \times 0.6$	$(0,\ 0,\ 0)$	–
Glowing border $\times 4$ (each side)	Box	$5.0 \times 0.08 \times 0.06$, $0.08 \times 1.9 \times 0.06$	$y = \pm 0.95$; $x = \pm 2.45$; $z = \pm 0.29$	–
Black screen (each side)	Box	$4.8 \times 1.8 \times 0.04$	$(0,\ 0,\ \pm 0.29)$	–
Digit bars (each digit)	Box × up to 7	across 0.61×0.11, up 0.11×0.65	digits at $x = \pm 1.65,\ \pm 0.60$	back side mirrored
Colon dots $\times 2$ (each side)	Box	$0.12 \times 0.12 \times 0.02$	$(0,\ \pm 0.25,\ \pm 0.33)$	–

8.7 The glass bridge

The bridge is entirely boxes. Step i ($i = 0..15$) is at $z_i = 3 + 3.3\,i$, at height 16. The full bridge is 54.3 long (z from 2.0 to 56.3, centre 29.15).

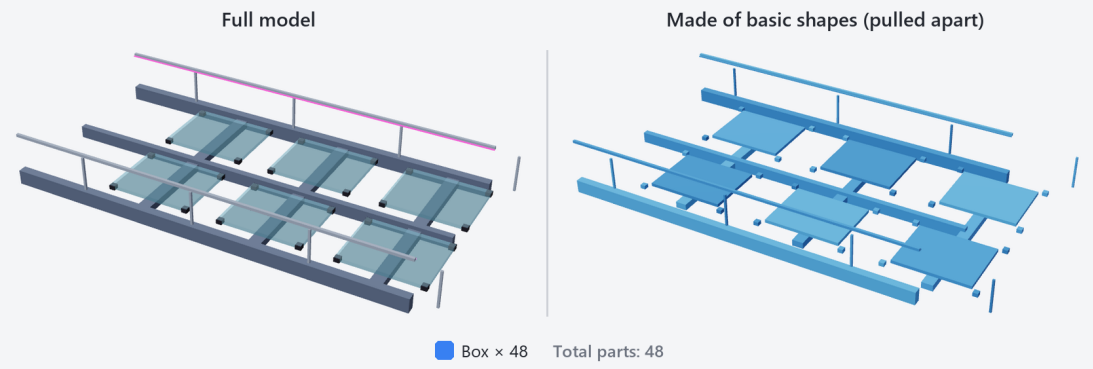


Figure 15: The first 3 steps of the bridge: 48 boxes (the full 16-step bridge uses 217).

Part	Shape	Scale / size	Translate (position)	Rotate
Steel beams $\times 2$	Box	$0.24 \times 0.42 \times 54.3$	$(\pm 2.75,\ 15.78,\ 29.15)$	–
Hand rails $\times 2$	Box	$0.08 \times 0.08 \times 54.3$	$(\pm 2.75,\ 17.05,\ 29.15)$	–
Pink neon strips $\times 2$	Box (glowing)	$0.03 \times 0.03 \times 54.3$	$(\pm 2.69,\ 17.02,\ 29.15)$	–
Rail posts $\times 34$	Box	$0.06 \times 1.05 \times 0.06$	$(\pm 2.75,\ 16.45,\ z_i - 1.65)$	–
Middle beam	Box	$0.18 \times 0.35 \times 54.3$	$(0,\ 15.78,\ 29.15)$	–
Cross beams $\times 16$	Box	$5.6 \times 0.20 \times 0.35$	$(0,\ 15.75,\ z_i)$	–
Glass panes $\times 32$	Box (42% see-through)	$1.90 \times 0.10 \times 2.40$	$(\pm 1.35,\ 15.98,\ z_i)$	–
Corner clamps $\times 128$	Box	$0.16 \times 0.12 \times 0.16$	pane $+(\pm 0.89,\ -0.06,\ \pm 1.13)$	–
“Safe” green glow	Box (glowing)	$1.82 \times 0.02 \times 2.30$	just above a proven pane	–
Broken pane marks	Box $\times 4$	$0.24 \times 0.02 \times 0.24$ (red)	at the four corners	–

8.8 Start platform and bridge guards

Start platform placed with: $T(0,\ 15.75,\ -4.5)$. The finish platform is the same idea at the other end ($z = 61$), without rails. The deck uses the concrete texture.

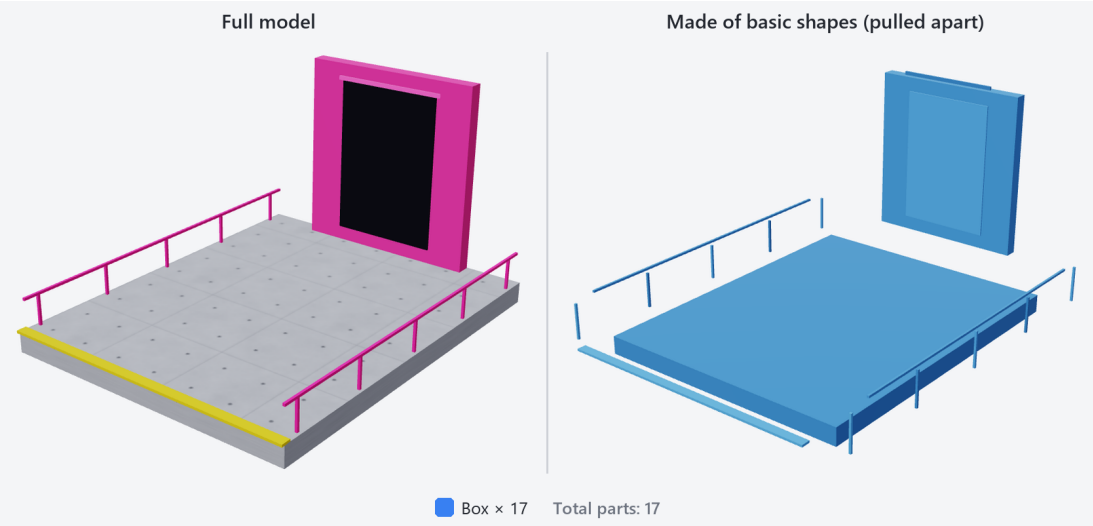


Figure 16: Start platform with its exit door: 17 boxes.

Part	Shape	Scale / size	Translate (position)	Rotate
Concrete deck	Box (textured)	$7.2 \times 0.5 \times 9.0$	$(0,\ 0,\ 0)$	–
Yellow edge	Box	$7.2 \times 0.06 \times 0.25$	$(0,\ 0.27,\ 4.45)$	–
Top rails $\times 2$	Box	$0.06 \times 0.06 \times 8.8$	$(\pm 3.5,\ 1.0,\ 0)$	–
Rail posts $\times 10$	Box	$0.06 \times 1.0 \times 0.06$	$(\pm 3.5,\ 0.5,\ -4, -2, 0, 2, 4)$	–
Door frame	Box	$4.4 \times 4.6 \times 0.35$	$(0,\ 2.5,\ -4.3)$	–
Dark doorway	Box	$2.6 \times 3.8 \times 0.42$	$(0,\ 2.5,\ -4.3)$	–
Glowing strip	Box (glowing)	$2.8 \times 0.10 \times 0.05$	door $+(0,\ 1.95,\ 0.22)$	–

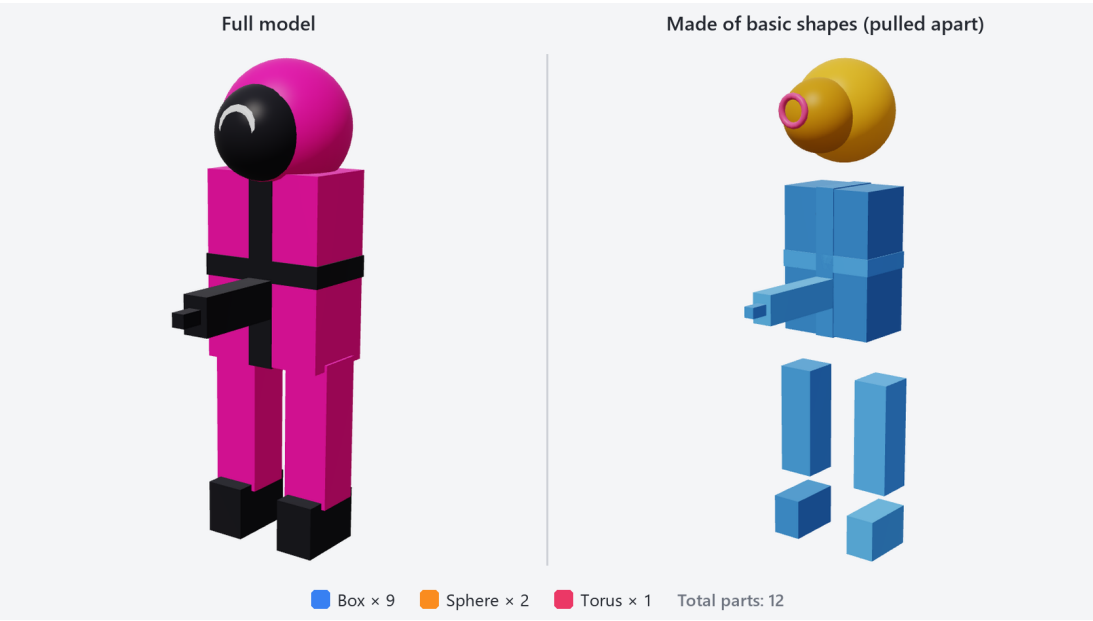


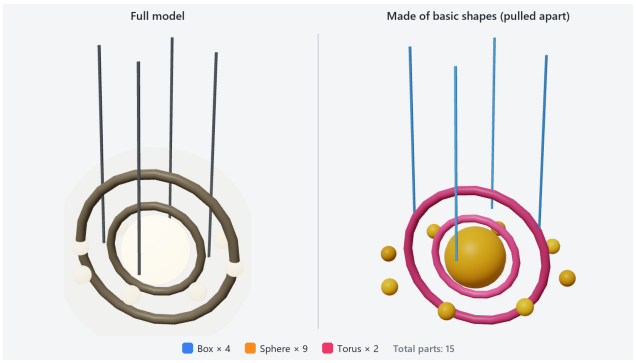
Figure 17: Bridge guard (simpler than the Round 1 guard): 9 boxes, 2 spheres, 1 torus.

Part	Shape	Scale / size	Translate (position)	Rotate
Boots $\times 2$	Box	$0.15 \times 0.20 \times 0.28$	$(\pm 0.16,\ 0.10,\ 0.02)$	–
Legs $\times 2$	Box	$0.18 \times 0.55 \times 0.18$	$(\pm 0.16,\ 0.45,\ 0)$	–
Body, belt, strap	Box $\times 3$	$0.48 \times 0.70 \times 0.30$; 0.49×0.08; 0.10×0.71	$(0,\ 1.05,\ 0)$	–
Rifle	Box $\times 2$	$0.10 \times 0.14 \times 0.52$; $0.05 \times 0.05 \times 0.70$	$(0,\ 0.95,\ 0.26)$	–
Head, mask	Sphere $\times 2$	radius 0.24; 0.17	$(0,\ 1.55,\ 0)$; $(0,\ 1.54,\ 0.14)$	–
Circle symbol	Torus	tube 0.015, ring 0.065	mask $+(0,\ 0.02,\ 0.14)$	–

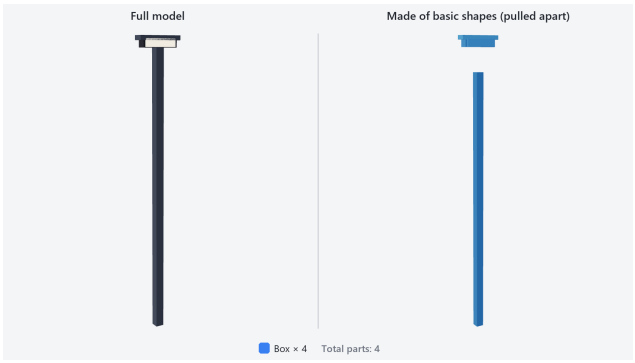
Placed with $T(x, 16, z) \cdot R_y(\text{angle})$: two guards at the start ($x = \pm 2.4$, angles $\pm 30°$) and two at the finish.

8.9 Light fixtures

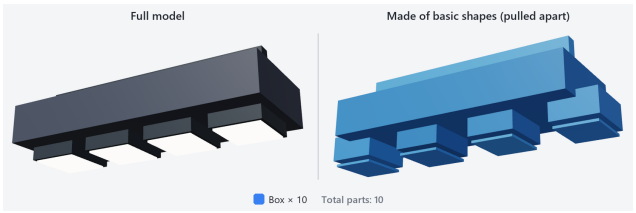
Each light in the game also has a visible model, so you can see where the light comes from. The glowing faces use *emission* (they light themselves, see Section 10).



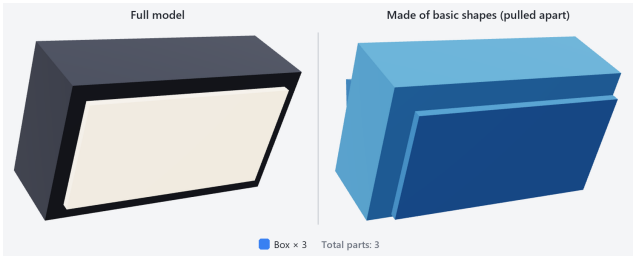
(a) Chandelier (point light, Round 2)



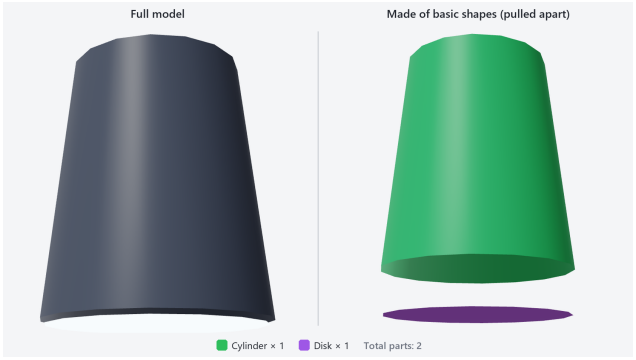
(b) Floodlight tower (point light, Round 1)



(c) Ceiling gantry (directional light, Round 2)



(d) Wall lamp (new point lights, Round 2)



(e) Follow spotlight (spot light, Round 2)

Figure 18: The light fixtures and their shapes.

Fixture	Parts (shape, size, translate, rotate)	Placed at
Floodlight tower	Box pole $0.45 \times 16 \times 0.45$ at $(0, 8, 0)$; Box bar $2.4 \times 0.25 \times 0.6$ at $(0, 16, 0)$; Box housing $2.0 \times 0.6 \times 0.5$ at $(0, 15.8, 0)$; glowing Box lamp $1.8 \times 0.45 \times 0.08$ at $(0, 15.75, -0.26)$	$T(18, 0, 15)$
Sun	glowing Sphere r 3.2 + 2 faint Sphere halos r 6.0, 10.5	$55 \cdot (\cos e \sin a,\ \sin e,\ \cos e \cos a)$, e=38°, a=45°
Chandelier	4 Box cables $0.06 \times 5.5 \times 0.06$ at $(1.5 \cos 90°k,\ 2.75,\ 1.5 \sin 90°k)$; 2 Torus rings (ring 2.2 and 1.3); 8 Sphere bulbs r 0.25 at $(2.2 \cos 45°i,\ -0.22,\ 2.2 \sin 45°i)$; glowing Sphere core r 0.95	$T(0, 23, 28)$
Follow spotlight	Cylinder body r 0.40 → 0.52, height 1.2, $R_x 90°$ (points down); glowing Disk lens r 0.5 at $(0, -1.22, 0)$	8.5 above the jumper
Ceiling gantry	Box rack $8.0 \times 1.2 \times 3.2$; Box hanger $7.2 \times 0.8 \times 0.4$ at $(0, 0.95, 0)$; 4 Box projectors $1.5 \times 0.8 \times 1.8$ at $(-3 + 2i,\ -0.6,\ 0)$, each with a glowing Box lens $1.35 \times 0.08 \times 1.6$	ceiling, y=28.2
Wall lamp $\times 4$	Box bracket $0.25 \times 0.25 \times 1.5$ at $(0, 0, -0.75)$; Box housing $1.6 \times 0.9 \times 0.5$, $R_x 25°$; glowing Box face $1.4 \times 0.7 \times 0.05$	$(\pm 18.4,\ 23.5,\ 14$ or $42)$, $R_y \mp 90°$

Table 3: Light fixture parts.

8.10 The two environments



Figure 19: Round 1 arena, normal and in shape view. The sky walls and sand floor are polygons, clouds are squashed spheres.

Part	Shape	Scale / size	Translate (position)	Rotate
Sky dome (new)	Polygon 32×10 quads	radius 190, pale horizon → deep blue top	centre $(0, 0, 0)$	–
Sky clouds (new) $\times 8$	5 Sphere each	squashed spheres, radius 6.5–9	150 away, 34–54 high	turned to face the arena
Sand floor	Polygon 24×32 quads (textured)	48×150, texture repeats every 6 units	$y = -0.05$	–
Painted sky walls $\times 4$	Polygon (quad)	48 or 150 long, 22 high, blue gradient	$x = \pm 24$, $z = \pm 75$	–
Wall clouds $\times 23$	6 Sphere each	$S(r,\ 0.65r,\ 0.08)$, $r = 1.7$–2.5	on the walls	–
Wheat and tree bands	Polygon $\times 8$	$y = 0$–3.2 and 3.2–4.2	just inside each wall	–
Start line	Box	$48 \times 0.01 \times 0.35$	$(0,\ 0.01,\ -52)$	–
Finish line	Box (glowing)	$48 \times 0.01 \times 0.45$	$(0,\ 0.01,\ 45)$	–



Figure 20: Round 2 hall, normal and in shape view.

Part	Shape	Scale / size	Translate (position)	Rotate
Side walls $\times 2$	Box (concrete)	$0.4 \times 48 \times 88$, texture 11×6 times	$(\pm 20,\ 10,\ 28)$	–
End walls $\times 2$	Box (concrete)	$40 \times 48 \times 0.4$	$(0,\ 10,\ -16)$, $(0,\ 10,\ 72)$	–
Ceiling, pit floor	Box (concrete)	$40 \times 0.4 \times 88$; $40 \times 0.5 \times 88$	$y = 34$; $y = -14$	–
Ceiling light strips $\times 2$	Box $\times 2$ each	$1.6 \times 0.25 \times 82$ + glowing panel	$(\pm 8,\ 33.7,\ 28)$	–
Wall ribs + LED bars $\times 22$	Box $\times 2$ each	$0.22 \times 48 \times 0.40$; $0.04 \times 6 \times 0.10$	$x = \pm 19.76$, $z = -9, -1, \ldots, 71$	–
Neon symbols $\times 6$	Torus or 3/4 Box + backing Box	ring radius 2.0; triangle side 4.2; square side 3.6	$(\pm 19.7,\ 25,\ 9 \mid 29 \mid 49)$	$R_y \mp 90°$; triangle sides $R_z \pm 30°$
Neon wall lines $\times 2$	Box (glowing)	$0.05 \times 0.14 \times 88$	$(\pm 19.78,\ 16,\ 28)$	–
Roof trusses $\times 5$	Box	$40 \times 0.6 \times 0.6$	$(0,\ 28.5,\ 0..56$ every $14)$	–
Hanging cables $\times 10$	Cylinder	radius 0.025, length 11.4	$(\pm 2.75,\ 17.1,$ truss $z)$	$R_x - 90°$
Safety net	Box $\times 39$	cables 0.03 thick	$y = 2$, 14×60 area	–

9 Lighting

The game uses all three classic light types. The shader adds up the light from every lamp for every pixel.

Light type	Idea	Round 1	Round 2
Directional	Far away like the sun. Same direction everywhere, no fading.	The sun (elevation $38°$, azimuth $45°$)	Ceiling gantry (elevation $50°$)
Point	Shines in all directions from one spot. Gets weaker with distance.	Floodlight tower at $(18, 16, 15)$	Chandelier at $(0, 23, 28)$ **+ 4 new wall lamps**
Spot	A point light limited to a cone, with a soft edge.	From the doll's eyes: red when she looks, green when she sings	Follows the jumping player from above

Table 4: The three light types in each round. Keys `0`, `8`, `9` turn them on and off.

How strong is the light at a point?

- **Diffuse** (matte brightness): $\max(N \cdot L,\ 0)$ – brightest when the surface faces the light.
- **Specular** (shiny spot, Blinn-Phong): $\max(N \cdot H,\ 0)^n$, where $H = \dfrac{L+V}{|L+V|}$ and n is the shininess (bigger n = smaller, sharper highlight).
- **Point light fading:** $f(d) = \dfrac{I}{1 + k_l\, d + k_q\, d^2}$ (tower: I=1.15, k_l=0.035, k_q=0.005; wall lamps: I=1.25, k_l=0.030, k_q=0.004).
- **Spot cone:** cone $= \text{clamp}\left(\dfrac{\cos\theta - \cos 23°}{\cos 15° - \cos 23°},\ 0,\ 1\right)$. Inside $15°$ the light is full, outside $23°$ it is zero, in between it fades.

Example. At distance $d = 10$ from the tower: $f = 1.15/(1 + 0.35 + 0.5) = 0.62$. So the light is about 62% as strong.

The pictures below were taken with only one light switched on at a time (the soft ambient light was lowered so each light is easy to see).

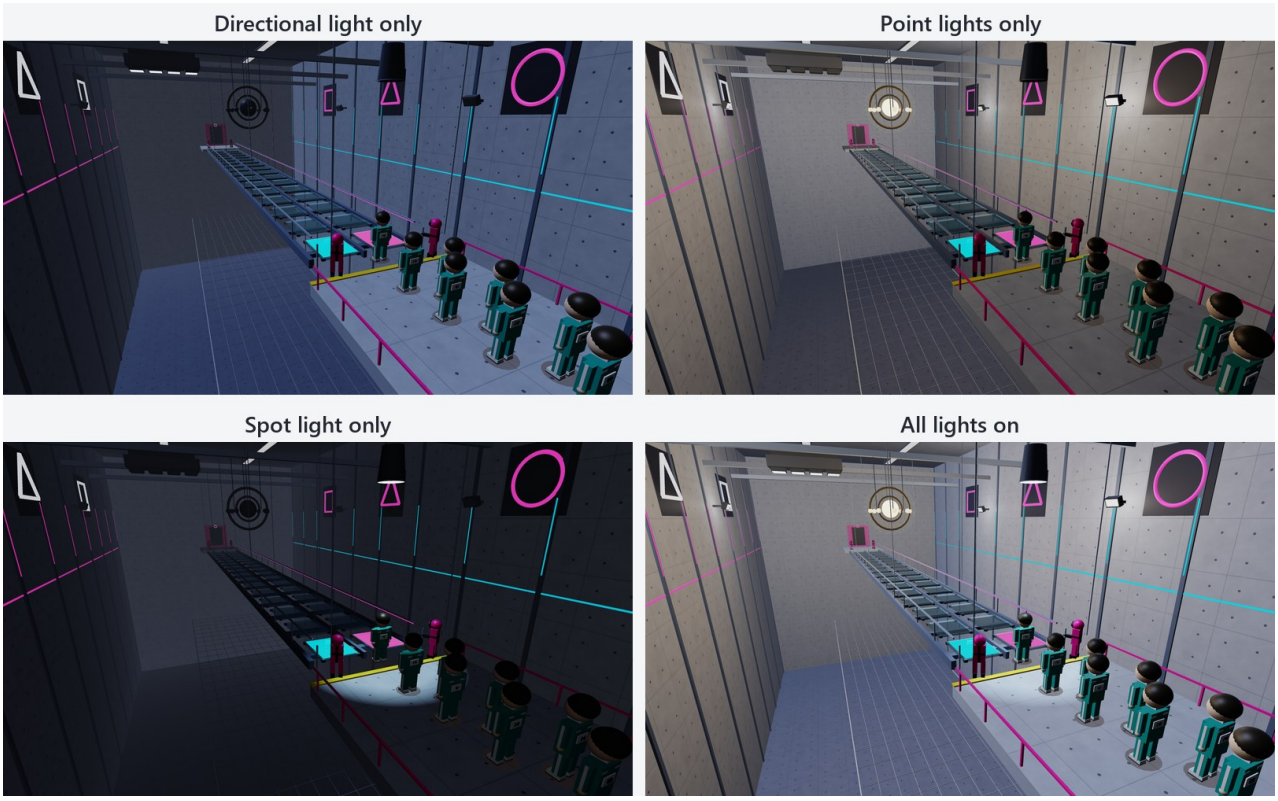


Figure 21: Round 2. Directional: even light from above. Point: warm light around the chandelier and the wall lamps. Spot: a bright circle on the player waiting at the front.

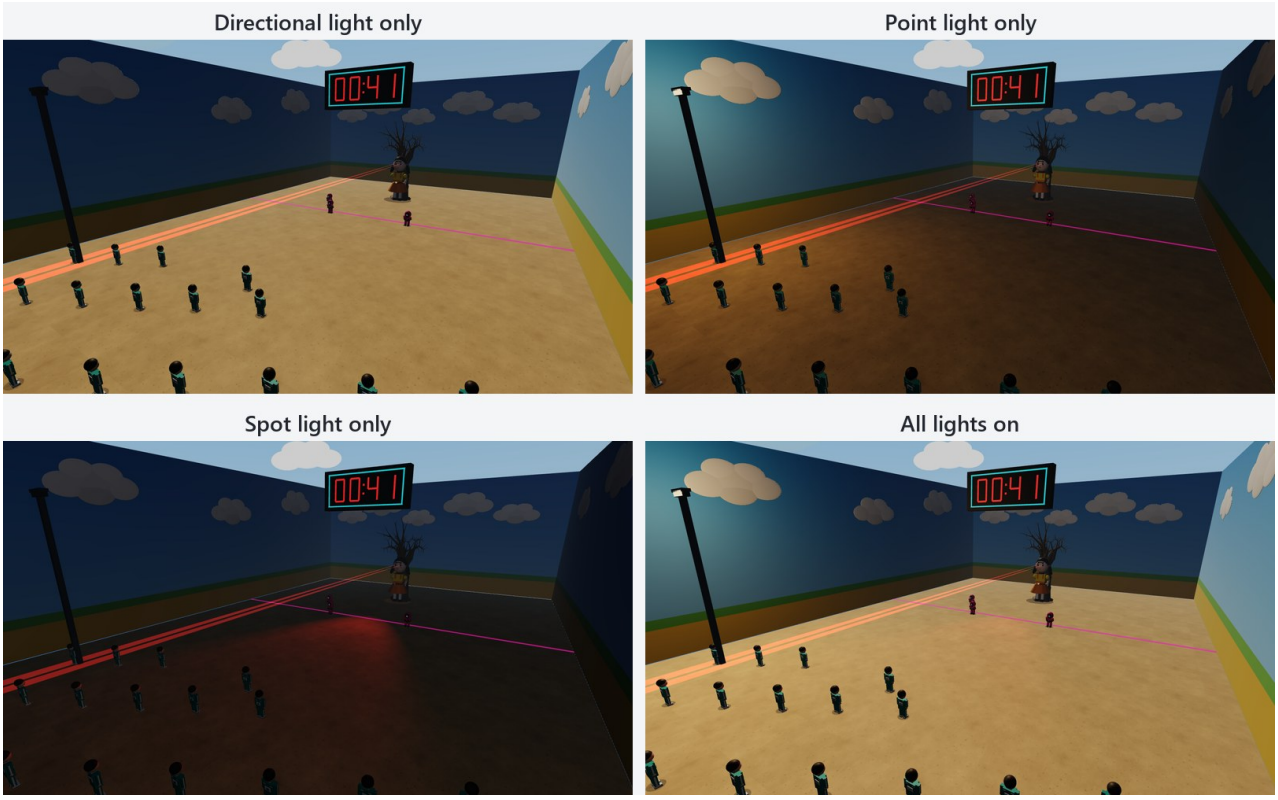


Figure 22: Round 1 (red light). Directional: sunlight on everything. Point: warm glow near the tower. Spot: a red cone from the doll's eyes lights the sand in front of her.

Extra light effects.

- **Ambient:** a soft base light so shadows are not pitch black. It mixes a sky color (from above) and a ground color (from below).
- **Emission:** glowing parts (neon, lamp faces, the clock digits) add their own color.
- **Haze:** far away surfaces slowly fade into a haze color, visibility $= e^{-(0.0075\,d)^2}$ in Round 2. This adds depth to big spaces.

10 Phong Shading

Shading decides the color of each pixel inside a triangle. This project uses **Phong shading**: the light is calculated *separately for every pixel*.

1. The **vertex shader** only moves each corner (vertex) into place and passes its normal on.
2. The GPU **interpolates** (blends) the normals across the triangle.
3. The **fragment shader re-normalizes** the normal for each pixel and computes ambient + diffuse + specular for every light.

$$\text{color} = \underbrace{k_a\, C_{\text{ambient}}}_{\text{ambient}} + \sum_{\text{lights}} \Big(\underbrace{\max(N\cdot L, 0)\, C_{\text{light}}}_{\text{diffuse}}\ \text{base} + \underbrace{k_s \max(N\cdot H, 0)^n\, C_{\text{light}}}_{\text{specular}} \Big) + \text{emission}$$

Because the normal is correct at every pixel, curved things look smooth and get small, sharp highlights even in the middle of a triangle – see the doll's eyes and lips and the players' heads.



(a) Shiny highlights on the doll's eyes, lips and hair. **(b)** Smooth light and highlights on round heads.

Figure 23: Phong shading, close up.

The heart of the fragment shader (`src/shaders/default.frag`):

```
// Blinn-Phong diffuse + specular of one light coming from direction L
void addLight(vec3 N, vec3 V, vec3 L, vec3 lightColor, float diffuseScale,
              float specScale, float specExp, inout vec3 diffuse, inout vec3 specular) {
    float diff = max(dot(N, L), 0.0); // Lambert: cos(angle to light)
    diffuse += diff * lightColor * diffuseScale;
    vec3 H = normalize(L + V); // halfway vector
    float spec = pow(max(dot(N, H), 0.0), specExp); // shininess = highlight size
    specular += spec * lightColor * specScale * materialSpecularStrength;
}
...
vec3 N = normalize(Normal); // per-pixel normal: this is Phong shading
vec3 V = normalize(viewPos - FragPos); // direction to the camera
```

11 Texture Mapping

A **texture** is an image wrapped onto a surface. This version adds three textures (on purpose, not too many). They were made by a small script (`tools/make_textures.py`) so they tile without seams.



Figure 24: The three texture images (512×512 PNG).

How it works.

1. Each vertex now has a **texture coordinate** (u, v) between 0 and 1 that says which point of the image belongs to it. Boxes get a $(0, 0)$–$(1, 1)$ square on each face, cylinders wrap u around and run v along the length, spheres use longitude/latitude.
2. The image is loaded with `stb_image` and uploaded once as an OpenGL texture with mipmaps and `GL_REPEAT`.
3. In the fragment shader: base $=$ material $\times$ texture$(u \cdot r_u,\ v \cdot r_v)$. The repeat counts (r_u, r_v) tile the image so it keeps a real size on big surfaces.

Texture	Used on	How it repeats
Sand	Round 1 floor (48×150)	$u = x/6,\ v = z/6$: one image per 6×6 units
Bark	Every branch of the dead tree	around: $\approx 4\times$radius times; along: length$/1.6$ times
Concrete	Round 2 walls, ceiling, pit floor and both platforms	one image per 8×8 units (e.g. side wall 11×6)

```
GLCompat::setTexture(&SceneTextures::concrete(), 88.0f / 8.0f, 48.0f / 8.0f); // 11 x 6 repeats
GLCompat::drawBox(0.4f, 48.0f, 88.0f); // side wall
GLCompat::clearTexture();
// fragment shader:
if (useTexture) baseColor *= texture(diffuseTexture, TexCoord * texRepeat).rgb;
```



(a) Sand floor **(b)** Bark on the tree **(c)** Concrete hall wall

Figure 25: The textures inside the game.

12 Animation and Game Logic

What moves	How	Formula
Walking legs and arms	Legs swing forward and back like a pendulum; arms swing the other way.	$\theta_{\text{leg}} = 28^\circ \sin(10\,t)$, $\theta_{\text{arm}} = -\theta_{\text{leg}}$
Doll's head	Turns smoothly to the target angle (0° = singing, 180° = looking).	angle $+=$ (target $-$ angle) $\cdot\, 15\,\Delta t$
Glass bridge jump	Moves straight to the next pane and rises in an arc.	$y = 16 + 1.3\sin(\pi p)$, $p : 0 \to 1$ in 0.42 s
Falling player	Speed grows every frame (gravity).	$v \mathrel{+}= 32\,\Delta t$, $y \mathrel{-}= v\,\Delta t$
Glass shards (55)	Random start speed and spin, then gravity.	$v_y \mathrel{-}= 28\,\Delta t$; $\vec{p} \mathrel{+}= \vec{v}\,\Delta t$
Blood drops (30)	Burst outward, fall, fade out.	$v_y \mathrel{-}= 9.8\,\Delta t$; alpha $= 1 -$ life/max

Round 1 rules.

- 60 seconds to reach the finish line ($z = 45$).
- Green light: the doll sings for 4.8–6.2 seconds. Red light: she looks for 3.2–5.2 seconds.
- If Player 456 moves while her head is turned past 80°, a guard shoots and the game is over.
- Each red light, every other contestant has an 18% chance to be caught.

Round 2 rules.

- 16 players take turns. Player 456 gets a random place in the queue.
- For every step, the strong side is chosen at random (50/50). Once a pane is proven, later players follow it.
- The other players choose randomly. Player 456 chooses with `A` / `D`. 16 minutes in total.

13 Cameras

The view matrix comes from `glm::lookAt(eye, target, up)` and the projection from `glm::perspective(fov, width/height, near, far)`. Using the real window width/height keeps round things round on any screen. Cameras glide to their new position smoothly: pos $+=$ (target $-$ pos) $\cdot\, (1 - e^{-k\,\Delta t})$.

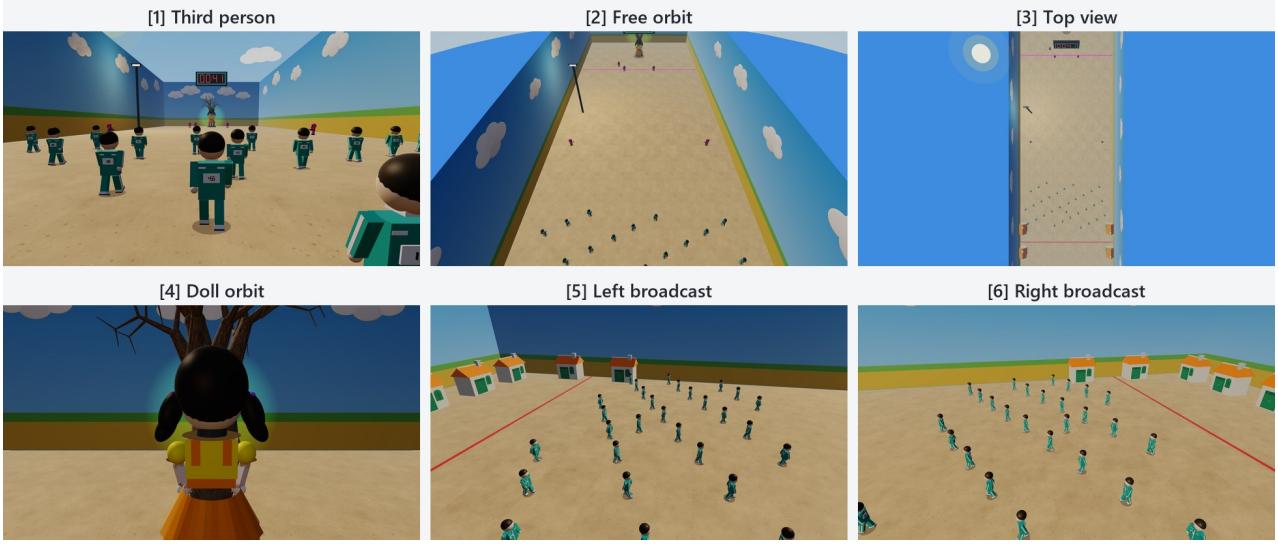


Figure 26: Round 1 camera modes (keys 1–6).

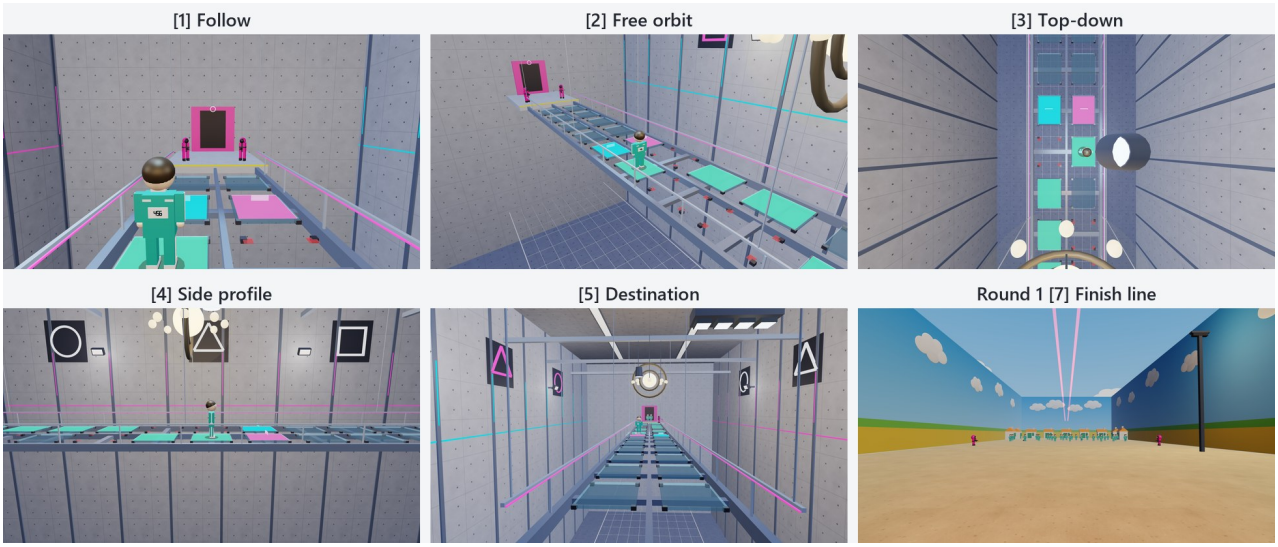


Figure 27: Round 2 camera modes (keys 1–5) and Round 1's finish-line camera (key 7, during red light).

14 What Changed in This Version

- **Brighter glass bridge hall.** Light concrete walls instead of near-black ones, four warm wall lamps (real point lights) on the two sides, two ceiling light strips, LED bars, big neon symbols and a soft haze. Ambient light raised from 0.22 to 0.50.
- **Phong shading only.** The old Gouraud mode and its key were removed; the shaders are shorter and clearer.
- **Three textures** (sand, bark, concrete) with proper texture coordinates on every basic shape.
- **Better outside look.** A sky dome with a color gradient and floating clouds above Round 1, 4× anti-aliasing (smooth edges), and a clean Windows font for the panels.
- **Shape view** (`I`) shows what every object is made of, live in the game.
- **Capture mode** (`--capture`) makes all screenshots in this report automatically.
- **Bugs fixed:**
  - The doll's red laser beams pointed backwards into the tree – now they point at the players.
  - The guard's triangle mask symbol was upside down (▽) – now it is △.
  - The doll camera picture (left panel) was hidden behind the panel – now it is drawn on top.
  - The finish door's ring badge was on the back of the door – moved to the front.
  - The top-down bridge camera was above the new ceiling – moved below it.
  - Pink blush on the doll's cheeks flickered (z-fighting) – made slightly thicker.
  - Symbols "●" and "₩" showed as "?" in the panels – fixed with the new font.
  - Old fixed-function calls (`glDisable(GL_LIGHTING)`, wide lines) that are errors in modern OpenGL were removed.

15 Conclusion

This project builds a complete two-round 3D game without loading a single 3D model. Seven basic shapes, placed with translate, rotate and scale and organised in parent–child groups, make every object – from a contestant (53 parts) to the giant doll (147 parts) and the recursive tree (313 parts). The scenes are lit by directional, point and spot lights, shaded per pixel with Phong shading, and three textures add surface detail. The new capture and shape-view tools make it easy to see, check and explain how everything is built.